

Simple Linear Regression with a Dirty Dataset

This notebook demonstrates:

- Creating a dirty dataset
- Cleaning the data
- Training a simple linear regression model
- Evaluating and visualizing results

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

Create a Dirty Dataset

```
data = {
    'hours': [1, 2, 3, 4, 5, 6, 7, 8, 'nine', 10, None, 12, 100],
    'score': [10, 20, 30, 40, 50, 60, None, 80, 90, 'hundred', 55, 120, 500]
}

df = pd.DataFrame(data)
print('Original Dirty Dataset:')
print(df)
```

Data Cleaning

```
# Convert to numeric
df['hours'] = pd.to_numeric(df['hours'], errors='coerce')
df['score'] = pd.to_numeric(df['score'], errors='coerce')

# Drop missing values
df = df.dropna()

# Remove outliers
df = df[(df['hours'] < 50) & (df['score'] < 200)]

print('\nCleaned Dataset:')
print(df)
```

Prepare Data

```
X = df[['hours']]
y = df['score']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Train Model

```
model = LinearRegression()
model.fit(X_train, y_train)
```

Predictions & Evaluation

```
y_pred = model.predict(X_test)

print('MSE:', mean_squared_error(y_test, y_pred))
print('R2 Score:', r2_score(y_test, y_pred))
```

Visualization

```
plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel('Hours Studied')
plt.ylabel('Score')
plt.title('Simple Linear Regression')
plt.show()
```

▼ Model Parameters

```
print('Slope:', model.coef_[0])
print('Intercept:', model.intercept_)

# Example prediction
new_hours = np.array([[9]])
print('Predicted score for 9 hours:', model.predict(new_hours)[0])
```